Physics of the Earth and Planetary Interiors, 64 (1990) 247–260
Elsevier Science Publishers B.V., Amsterdam

Palaeomagnetism of ocean layers 2 and 3: evidence from the Metchosin Complex, Vancouver Island

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(Accepted for publication December 5, 1988)

ABSTRACT

Irving, E. and Massey, N.W.D., 1990. Palaeomagnetism of ocean layers 2 and 3: evidence from the Metchosin Complex, Vancouver Island. Phys. Earth Planet. Inter., 64: 247–260.

The early Eocene Metchosin Complex is an ophiolitic sequence and is the northernmost part of the Crescent Terrane. It was underthrust beneath Wrangellia later in the Eocene. Directions of magnetization in sheeted diabase (layer 2) and gabbroic intrusions (layer 3) from the complex are internally consistent, and both normal and reversed polarities are present. Mean directions, irrespective of sign, differ not only from the expected Eocene palaeofield but also from one another. These divergences have not, therefore, been caused by tectonic rotation of the Metchosin Complex as a whole, but by variable tilting of bodies within the complex. It is argued that these tilts probably were imposed during the emplacement of the Metchosin Complex adjacent to and beneath Wrangellia. The systematic clockwise rotations found further south in Eocene rocks of the Crescent Terrane of Washington and Oregon do not occur. The remanent magnetizations of the gabbros of layer 3 are commonly strong and stable. These are the only rocks in the Metchosin Complex that, had they been part of the present oceanic crust, would be capable of contributing to marine magnetic anomalies. The stable magnetization of diabase of layer 2 is too weak, the dominant magnetization being a soft viscous remanence imposed by the present geomagnetic field. The remanence of basalt lavas of layer 2, as previous work on the Metchosin Complex has shown, is weak and scattered in direction.

1. Introduction

The early Eocene Metchosin Complex occurs on southernmost Vancouver Island immediately west of Victoria (Fig. 1). It is the northerly continuation of the Coast Range basalts of western Washington and Oregon, a Palaeocene to Eocene igneous sequence of oceanic origin. The Coast Range basalts form the floor of the Crescent Terrane. Formerly the Metchosin Complex was believed to consist of two units, Eocene Metchosin volcanics and Oligocene gabbroic stocks (Clapp and Cook, 1917; Muller, 1980). Recently obtained field evidence indicates that the Metchosin Complex is ophiolitic (Massey, 1986). Of particular importance is the identification of sheeted dykes, and the recognition that the gabbroic intrusions are not younger intrusions but are coeval and stratigraphically part of the same sequence (Fig. 2). These sheeted dykes and gabbros provide the opportunity to study the magnetic properties of ocean layers 2B and 3, which are of interest because the contribution that their magnetizations make to marine magnetic anomalies is not yet fully understood.

The remanent magnetism of Metchosin rocks was first studied by Cox in 1958 (A. Cox, private communication, 1979). Earlier, Cox (1957) had observed a large easterly declination in the Eocene Siletz volcanics of Oregon, which was later inter-

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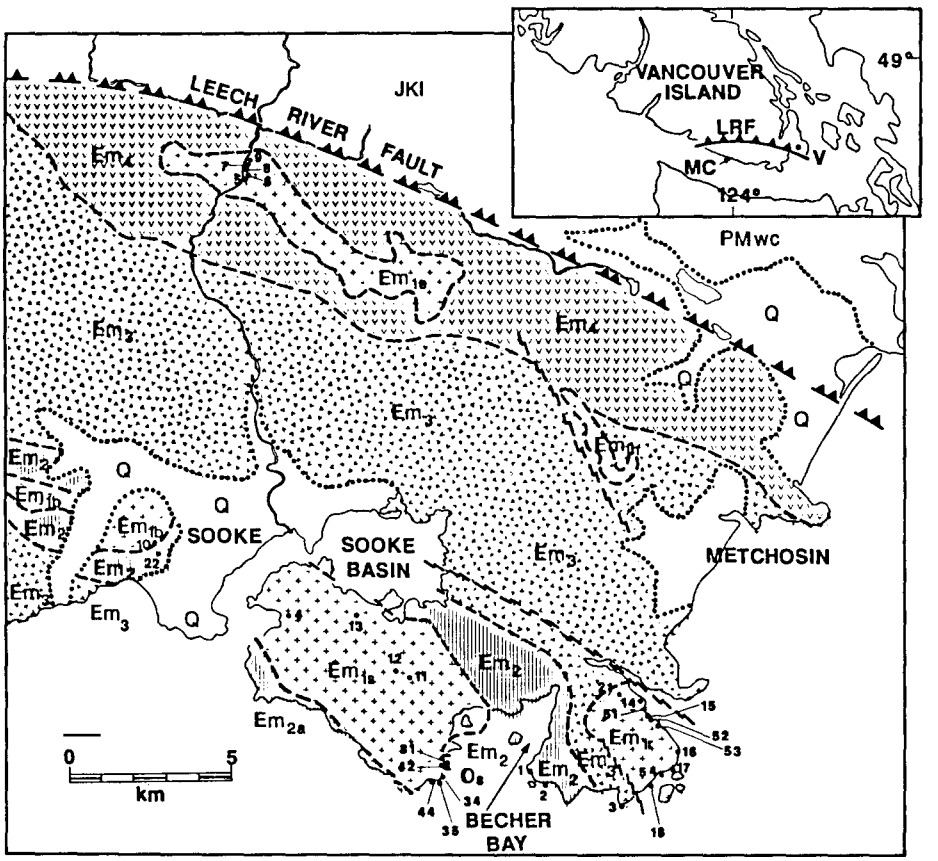


Fig. 1. Geology and sampling. Em_{1b} = Broom Hill stock; Em_{1s} = East Sooke stock; Em_{1r} = Rocky Point stock; Em_{1e} = Empress Mountain sill; Em_{1h} = Mount Helmecken stock; Em_2 = sheeted dykes; Em_{2a} = sheeted dykes within agmatitic intrusive breccia of East Sooke stock; Em_3 = submarine volcanics; Em_4 = subaerial massive flows; Os = Sooke Formation; JK1 = Leech River complex; PM wc = Wark–Colquitz gneiss; Q = Quaternary deposits. Location of main map area shown in the inset: LRF = Leech River fault; MC = Metchosin Complex; V = Victoria.

preted as a 63° clockwise tectonic rotation of the Coast Range (Irving, 1964; Cox, 1980). Cox wanted to know if this anomalous declination extended along the length of the Coast Range. He sampled pillow lavas, found them magnetically unstable and abandoned the work. In 1968, E. Irving, unaware of Cox's work, resampled the same pillow lavas, again found them unstable and likewise abandoned the work.

Then Symons (1973) reported data from the East Sooke gabbroic stock. He found magnetizations with normal and reversed polarity, which were stable under alternating field (a.f.) demagnetization. The magnetization directions differed from those expected from cratonic North America. This discordance could have been caused by tilting of the East Sooke stock or by rotating about a vertical axis. Symons argued that tilting was improbable and that "the discordance is caused by a 20° counterclockwise twist rotation of the Tertiary belt at the southern end of Vancouver Island relative to the rest of the island to the north and to cratonic North America." This was in conflict with the clockwise rotations that are characteristic of the Crescent Terrane in Washington and Oregon (Grommé et al., 1986). The main purpose of this investigation is to determine if the aberrant magnetizations observed by Symons truly reflect an anticlockwise rotation of the Metchosin Complex as a whole, or if they are the result of local tilting;

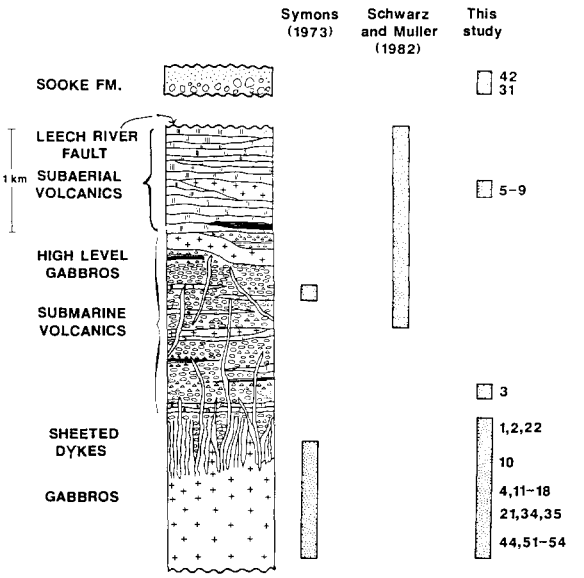


Fig. 2. Stratigraphic column modified from Massey (1986) showing the distribution of previous and present (with site numbers) palaeomagnetic sampling.

if all bodies show the same aberrancy as observed by Symons then wholesale rotation is probable, if the magnetizations are not coherent from body to body, then local tilting is the likely cause.

Symons (1973) also studied a few specimens of Metchosin lava and found them to possess soft magnetizations directed along the present Earth's magnetic field, but there was also present a weak reversed magnetization essentially parallel, when no correction is made for geological tilt, to that he had observed in the nearby East Sooke stock. He interpreted the latter as a thermoremanent magnetization acquired when the stock was intruded.

More recently, Schwarz and Muller (1982) studied the Metchosin lavas at nine sites (Fig. 2). After cleaning at certain alternating fields (a.f.) and temperatures, the authors claimed to have isolated a magnetization (D, I, $= 74°$, $-66°$, $\alpha_{99} = 14°$) at three sites. This direction is strongly divergent from that expected from the craton. However, the magnetizations they describe are very complex, and the technique of single-step cleaning is probably not applicable. The directions were generally scattered, and only one site showed good internal coherence. Of significance, we argue, is the fact that after a.f. cleaning a proportion of the directions in situ were clustered around the normal direction of magnetization of the East Sooke stock, the remainder being randomly distributed (Schwarz and Muller, 1982, fig. 11.1). Therefore, we would extend the argument of Symons (1973) and interpret the former as being an overprint acquired during intrusion of the stock, and the latter as noise of no significance. The intensity of natural remanent magnetization (NRM) recorded by Schwarz and Muller was 10^{-1} A m^{-1}, an order of magnitude too low to produce marine magnetic anomalies.

2. Geology and sampling

The general structural relationships of the Metchosin Complex are shown in Fig. 3. It is fault bounded. To the north it has been thrust beneath older Mesozoic and Palaeozoic rocks of Wrangellia along the Leech River fault (Yorath et al., 1985; Clowes et al., 1987). The base of the complex is not exposed on Vancouver Island, but interpretations of LITHOPROBE seismic reflection profiles suggest that it also is a northward-dipping thrust that lies at a depth of ~ 6–8 km, and which could be the northward continuation of the Hurricane Ridge fault exposed to the south in the Olympic Mountains of northwest Washington (Clowes et al., 1987). The complex is cut by high-angle faults with modest displacements. Local dips within the complex are variable. As a whole, however, the complex dips ~ 30° to the northeast, but further south, toward the Clallam syncline, southerly dips are to be expected. During formation, lava flows and dykes have suffered low-grade metamorphism in the zeolite and prehnite–

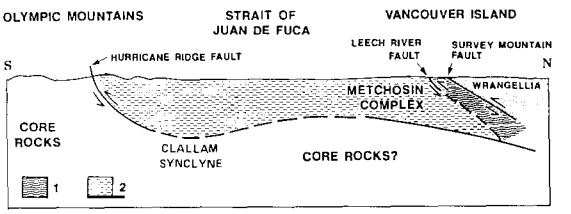


Fig. 3. General tectonic setting of Metchosin Complex adapted from Clowes et al. (1987): (1) Leech River schists; (2) Eocene oceanic rocks.

TABLE 1

Ages from the Metchosin Complex

Rock type	Sample	Method	Age (Ma, $\pm 1\sigma$)	Reference
Geochronometric data				
Volcanics				
Basalt flow	D80-MT-2 (i)	WR, Ar–Ar fusion	57.8 ± 0.8	Duncan (1982)
	(ii)	WR, K–Ar	53.1 ± 0.7 [a]	Duncan (1982)
Basalt flow	D80-MT-3	WR, K–Ar	43.3 ± 0.8 [a]	Duncan (1982)
Basalt flow	D80-MT-4	WR, K–Ar	41.9 ± 0.9 [a]	Duncan (1982)
Gneissic metavolcanic		Hb, K–Ar	46.9 ± 7	Muller (1977)
Gabbro stocks				
Hornblende diorite	345-71	Zr, U–Pb	52 ± 2	R. Zartman [b]
Quartz diorite	RWT-345-71	Hb, K–Ar	54.6 ± 1.3 [a]	Marvin and Cole (1978)
Diorite	RWT-344-71	Hb, K–Ar	47.5 ± 3.1 [a]	Marvin and Cole (1978)
Uralitized gabbro	RWT-347-71	Hb, K–Ar	47.4 ± 1.7 [a]	Marvin and Cole (1978)
Uralitized gabbro	RWT-348B-71	Hb, K–Ar	39.2 ± 1.6 [a]	Marvin and Cole (1978)
Amphibolite inclusion (intrusive breccia)	NWM 843001	Hb, K–Ar	52.2 ± 4.5	Massey and Parrish [c]
Palaeontological data				
Metchosin Volcanics				
Turritella in interbedded sediment			Probably early Eocene	W.O. Addicott [d]
Interflow bioclastic limestone			Early Eocene	B.E.B. Cameron [e]

[a] Ar loss suspected or age regarded as minimum by reporter.
[b] Unpublished data, written communication to M.T. Brandon.
[c] Unpublished data (error is $\pm 2\sigma$).
[d] In Muller (1977).
[e] Personal communication (1984).

pumpellyite facies, resulting in the deposition of chlorite, albite, epidote, pumpellyite, laumontite and stilbite in veins and vesicles, and disseminated through the rock. Conglomerate and sandstone of the Upper Oligocene Sooke Formation unconformably overlie the complex, filling erosional depressions.

Unlike other sequences of the Crescent Terrane, the Metchosin Complex exhibits an ophiolitic pseudostratigraphy (Massey, 1986). Isotropic and layered gabbro and leucogabbro form several stocks predominantly at the base of the exposed sequence. Some gabbro bodies occur higher in the section, where they have intruded volcanic rocks (Fig. 2). Sheeted dykes occur marginal to the East Sooke and Broom Hill stocks (Fig. 3). The dykes are predominantly fine- to medium-grained diabase, but feldspar-diabase, microgabbro and basalt varieties are also present. Late-stage trondhjemite, differentiates from the gabbro, intrude the sheeted dykes. Basaltic volcanics, the Metchosin Volcanics of previous authors (Clapp and Cooke, 1917; Muller, 1980), structurally overlie the sheeted dykes. They can be subdivided into a lower submarine unit and an upper subaerial unit.

Ages are summarized in Table 1. Fossils, and zircon U–Pb and Ar–Ar (52 and 58 Ma) ages, which are regarded as the most reliable radiometric determinations, indicate an early Eocene age.

Sampling was concentrated in the gabbro stocks and sheeted dykes in the lower part of the complex (Figs. 1 and 2 and Table 2). Also studied were the Empress Mountain stock, which is intruded at a higher level into volcanics, and sandstones and conglomerates of the Sooke Formation, which overlies the East Sooke stock near Creyke Point. At each site, five to seven cores were taken.

TABLE 2

Data by sites [a]

Number	Latitude (N)	Longitude (W)	Rock type	C (n)	M_R	M_I	D, I (degrees)	k	α_{95} (degrees)	Treatment
East Sooke stock										
04	48°21.8′	123°42.0′	Leucogabbro	5 (08)	0.23	0.03	189, −63	69	7	50 mT, 100–650 °C
11	48°20.6′	123°39.1′	Gabbro	5 (12)	2.54	0.08	335, 63	107	4	30–80 mT, 150–700 °C
12	48°20.7′	123°39.3′	Gabbro	5 (07)	1.31	0.07	334, 59	147	5	30–80 mT, 150–700 °C
13	48°20.6′	123°40.4′	Olivine gabbro	5 (09)	0.13	0.08	154, −66	34	9	500–650 °C
34	48°19.0′	123°38.4′	Gabbro	6 (12)	2.57	0.13	311, 63	64	6	10–60 mT
35	48°18.9′	123°38.3′	Gabbro	6 (11)	3.15	0.06	329, 63	124	4	50 mT, 200–600 °C
44	48°19.0′	123°38.4′	Gabbro	6 (12)	2.57	0.07	311, 63	64	6	10–60 mT
Empress Mountain stock										
05	48°29.0′	123°42.9′	Diabase, gabbro	7 (13)	0.71	0.22	002, 78	67	5	10–50 mT, 100–650 °C
06	48°29.1′	123°42.9′	Coarse gabbro	5 (11)	3.14	0.75	213, 81	12	14	10–50 mT, 400–600 °C
07	48°29.1′	123°42.8′	Porphyritic gabbro	5 (12)	0.43	0.08	326, 78	91	5	10–50 mT, 100–650 °C
08	48°29.2′	123°42.8′	Tonalite	6 (13)	0.91	0.25	320, 76	90	4	10–50 mT, 100–650 °C
09	48°29.2′	123°42.8′	Pegmatitic gabbro	5 (09)	3.61	1.15	348, 77	45	8	10–60 mT, 100–650 °C
Rocky Point stock										
14	48°20.3′	123°33.2′	Gabbro	5 (08)	0.01	0.01	197, −38	34	9	30–80 mT, 150–650 °C
15	48°20.1′	123°33.1′	Gabbro	5 (11)	0.18	0.22	212, −30	52	6	30–100 mT
16	48°19.5′	123°32.4′	Leucogabbro	6 (14)	0.27	0.07	194, −39	83	4	80 mT
17	48°19.2′	123°32.5	Leucogabbro	5 (16)	1.33	0.04	168, −55	26	8	30–80 mT, 150–700 °C
18	48°19.0′	123°33.0′	Gabbro	6 (10)	0.40	0.02	002, 44	86	5	30–80 mT, 300–650 °C
21	48°20.3′	123°33.9′	Olivine gabbro	7 (15)	0.05	0.02	203, −32	66	5	100 mT, 300–700 °C
51	48°20.2′	123°33.1′	Gabbro	5 (10)	0.04	0.01	224, −09	55	7	60 mT
52	48°20.0′	123°30.0′	Gabbro	4 (07)	0.06	0.01	192, −10	83	7	60 mT
53N	48°19.5′	123°32.7′	Altered gabbro	4 (07)	0.06	0.01	348, 42	27	12	20 mT
53R	48°19.5′	123°32.7′	Altered gabbro	2 (03)	0.01	0.01	214, −3	41	20	60 mT
54	48°19.2′	123°32.7′	Gabbro	5 (10)	0.49	0.03	034, 49	153	4	60 mT
Broom Hill stock										
10	48°23.9′	123°45.2′	Gabbro, agmatite	25 (38)	0.30	0.02	169, −71	48	3	20–10 mT, 100–650 °C
Sheeted dykes (30° at 342° [d])										
01	48°19.2′	123°36.1′	Microgabbro	5 (08)	1.24	0.55	158, −24 (150, −54)	195	4	20–60 mT, 400–700 °C
02	48°19.0′	123°35.7′	Feldspar diabase	9 (12)	0.51	0.48	143, −31 (130, −58)	49	7	10–90 mT, 400–650 °C
22	48°23.9′	123°45.2′	Diabase	4 (07)	0.14	0.16	144, −67	47	9	20–80 mT, 550–600 °C
Christopher Point sill (30° at 342° [d])										
03	48°18.6′	123°33.7′	Gabbro	8 (13)	0.61	0.12	355, 56 (047, 82)	85	5	10–70 mT, 100–600°
Sooke Formation										
42	48°19.4′	123°38.0′	Sandstone	16 (28)	0.02		028, 71	41	4	10–60 mT
31	48°19.4′	123°38.0′	Conglomerate	16 (64)			Random	1	–	10–100 mT, 100–600 °C

[a] C is the number of cores; n is the number of specimens that are given weight, except in entry 31 where unit weight is given to 16 boulders (see Fig. 19); M_R is the intensity in A m^{-1}; M_I is the magnetization induced in a field of 0.1 mT (1 Oe); D and I are the declination and inclination of the mean of n specimens; k the precision, and α_{95} the error ($P = 0.05$).

[b] Directions of magnetization have been obtained either by treatment (cleaning) at the stated level, or by calculating the magnetization removed in the stated intervals using least-squares fit of line on orthogonal plots.

[c] Site 6 of Table 3 has not been used in the analysis of Table 2 because the error is too large.

[d] Attitude determined from adjacent volcanics.

Generally two cylindrical specimens were cut from each core.

3. Gabbros

Four gabbro bodies were studied. Thermal demagnetization curves for gabbro of the East Sooke stock (seven sites, Fig. 1) are square-shouldered, with unblocking temperatures mainly between 550 and 600° C. The directions remain unchanged up to 600° C (Fig. 4). Alternating field demagnetization typically yields linear decay, and a substantial proportion of the remanence is left after 100 mT. The directions remain unchanged throughout (Fig. 5). The magnetizations are single component and specimens have been cleaned at the a.f. and temperatures listed in Table 1. Mean directions for each site, together with previous data of Symons (1973) are plotted in Fig. 6. The means of normal and reversed groups are accurately opposed in direction (Table 3). The directions differ systematically from the expected Eocene field direction ('E' of Fig. 6), the inclinations, on average, being somewhat shallower and the declination displaced in an anticlockwise sense. Symons' result is fully confirmed, with the added observation that the unblocking temperatures are always very high and confined within a very narrow temperature interval (Fig. 4). These gabbros are ideal recorders of the geomagnetic field.

Thermal and a.f. demagnetization results for the Rocky Point stock (sampled at 10 sites, Fig. 1) are very similar to those observed in the East Sooke stock. Unblocking temperatures are confined in the narrow interval 550–600° C (Fig. 7), and 20% or so of the remanence is retained after treatment in 100 mT (Fig. 8). Normal and reversed magnetizations are present, and at one site (53, the only site at which the gabbro is extensively altered) both polarities are present. Mean directions, calculated in various ways and including one site sampled by Symons, are given in Table 3. In contrast to the East Sooke gabbro,

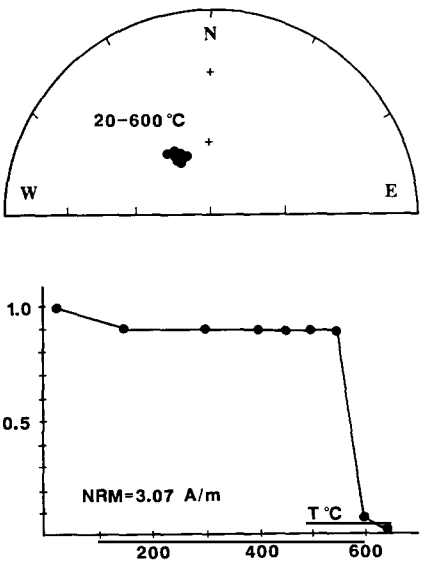


Fig. 4. Thermal demagnetization of East Sooke stock, site 11. Directions plotted on equal-area projection with solid (open) symbols denoting downward (upward) inclinations. Perimeter is present horizontal plane. Intensity changes below.

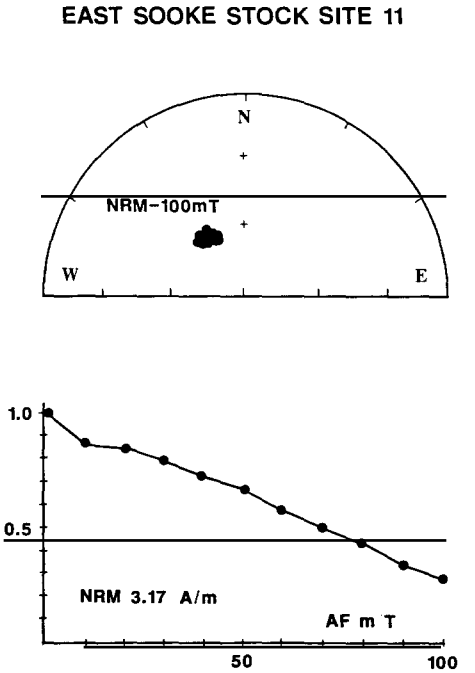


Fig. 5. Alternating field demagnetization of East Sooke stock, site 11. See caption to Fig. 4.

TABLE 3

Summary of data [a]

Rock unit	N	Pole	D_m, I_m (degrees)	k	α_{95} (degrees)	λ, ϕ (degrees)	K	A_{95} (degrees)
East Sooke stock (48.4° N, 123.7° W)								
(1) Present study	07	5N, 2R	332.0, 63.8	86	6.5	71.7, 152.4	36	10.2
(2) Normal, Symons' (1973) and present study	20	All N	327.4, 60.4	46	4.9	66.7, 144.2		
(3) Reversed, Symons' (1973) and present study	15	All R	157.2, −67.9	66	4.7	74.5, 170.9	28	7.4
(4) All data (2 + 3)	35	20N, 15R	330.9, 63.7	47	3.6	70.5, 153.2	23	5.2
Empress Hill stock								
(5) (48.5° N, 123.7° W)	04	All N	338.4, 77.8	350	4.9	69.1, 211.2	96	9.4
Broom Hill stock								
(6) (48.4° N, 123.8° W)	25 [b]	All R	166.0, −70.2	36	4.0	79.5, 185.7	–	6.5
Rocky Point stock (48.3° N, 123.5° W)								
(7) Present study excluding 53R	10	3N, 7R	017.0, 36.2	15	13.0	59.8, 025.6	17	12.2
(8) Entry 7 excluding 51, 52, 53R	08	3N, 5R	013.4, 42.3	28	10.7	64.4, 028.8	22	12.1
(9) Entry 9 adding Symons' site 35	11	3N, 8R	017.9, 34.8	16	12.0	60.5, 028.7	17	11.4
(10) Entry 9 excluding sites 51, 52	09	3N, 6R	015.0, 40.1	25	10.5	64.7, 033.0	23	11.0
Sheeted dykes (48.3° N, 123.6° W)								
(11) Sites 01, 02 with respect to present horizontal	20 [b]	All R	144.0, −57.8	25	6.6		–	5.7
(12) Sites 01, 02 with respect to layering (30° at 342°)	20 [b]	All R	150.9, −28.6	25	6.6	49.2, 102.3	–	5.7
(13) Sites 01, 02, 22 with respect to present horizontal	18 [b]	All R	144.0, −59.7	30	6.4	62.7, 143.7	–	8.4

[a] N is the number of sampling sites, except for entries 6, 11, 12 and 13; D_m and I_m are the declination and inclination of the mean with respect to the present horizontal, except for entry 12; $k(K)$ is the precision of directions (of site poles); λ and ϕ are the latitude (N) and longitude (E) of the palaeopole; $\alpha_{95}(A_{95})$ is the error in mean direction (in palaeopole).

[b] Number of cores.

declinations are systematically clockwise of the expected Eocene field direction ('E' of Fig. 9), and inclinations are shallower.

The Broom Hill stock and associated diabase (see below) were sampled over a distance of 600 m in one extended outcrop (Fig. 3). Once again the stability is high, the unblocking temperatures extending above 600 °C, indicating the presence of both magnetite and haematite (Fig. 10). Alternating field demagnetization yields exponential decay, 10% of the magnetization being retained at 100 mT (Fig. 11). The directions change little during demagnetization, are all reversed, and cluster about the expected Eocene field direction (Fig. 12, lower stereogram).

The Empress Mountain sill is intruded into basalt flows and is stratigraphically higher than the gabbros just described (Fig. 2). It was sampled at five sites along Boneyard Creek (Fig. 3). Once again, thermal decay curves are square-shouldered, somewhat less so than for other gabbroic bodies, with unblocking temperatures mainly between 600 and 650 °C, higher than in previous bodies (Fig. 13). The directions are all of normal polarity, and cluster near to the expected Eocene direction, but with systematically steeper inclinations (Fig. 12).

The Christopher Point sill (site 3) is also a high-level intrusion into pillow lavas, and is ~ 10 m thick. Unblocking temperature (400 °C) and coercivities of remanence (< 500 mT) are much lower than those of other gabbros. The directions, which are all normal and not unlike those of the

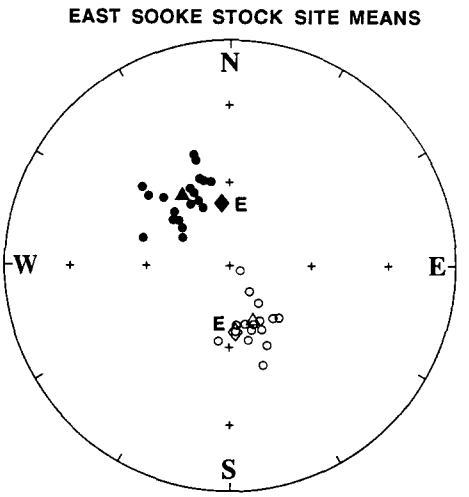


Fig. 6. Site-mean directions East Sooke stock. Of 35 site-mean directions, 28 are from Symons (1973) excluding those for which (α_{95}) exceeds 10°, and seven are from the present study (Table 2). Triangles are the mean directions of the normal (solid dots) and reversed (open circles) sites. E is the mean direction (normal and reversed diamonds) of the early Eocene field (~ 50 Ma) expected in the sampling area calculated from a mean palaeopole 85° N, 176° E ($A_{95} = 6°$) derived from the work of Lovlie and Opdyke (1974), Diehl et al. (1983) and Shive and Pruss (1977), as summarized in Bardoux and Irving (1988). The expected palaeofield direction is 353°, 68° ($\alpha_{95} = 5°$) and its opposite. Perimeter is the present horizontal.

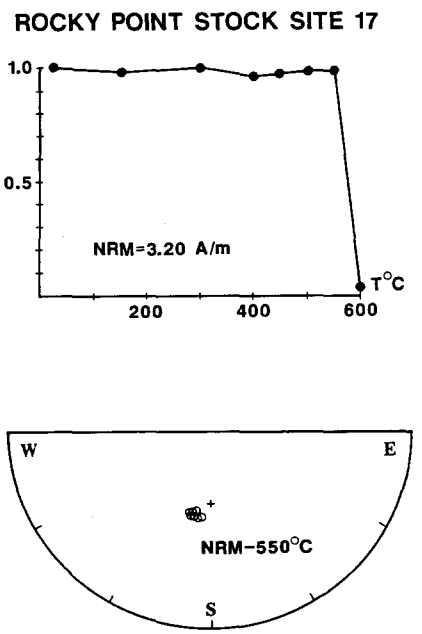


Fig. 7. Thermal demagnetization of Rocky Point stock site 17. See caption to Fig. 4.

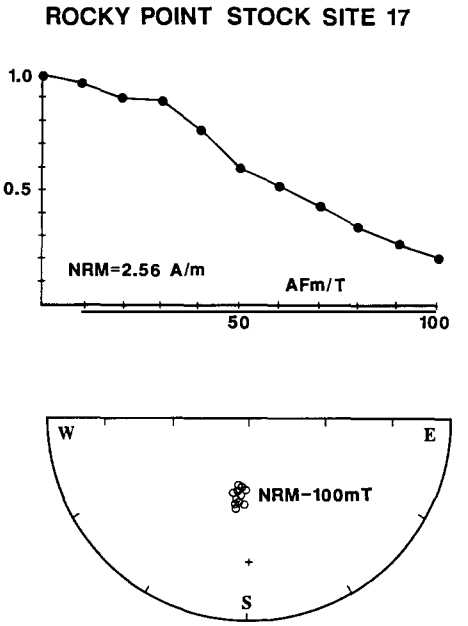


Fig. 8. Alternating field demagnetization of Rocky Point stock site 17. See caption to Fig. 4.

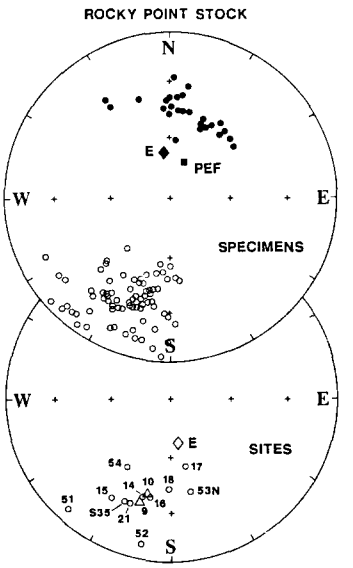


Fig. 9. Rocky Point stock. Specimen directions above, solid (open) symbols denote positive (negative) inclination. Below, site directions with normal polarity are inverted. E is the expected palaeofield direction for early Eocene time in the sampling area, normal above and reversed below. PEF is the direction of the present Earth's field in the sampling area. S35 is from Symons (1973). Triangles are mean directions corresponding to entries 9 and 10 in Table 3. Perimeter is the present horizontal.

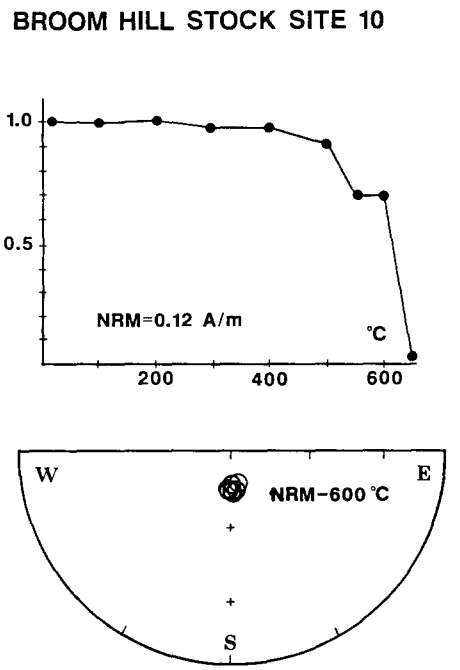


Fig. 10. Thermal demagnetization of agamatite from Broom Hill stock, site 10. See caption to Fig. 4.

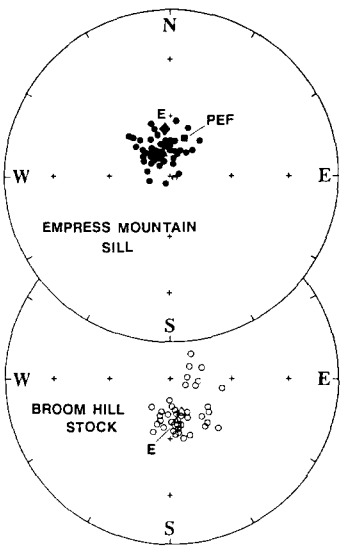


Fig. 12. Specimen directions from Empress Mountain sill above, Broom Hill stock below. Triangles are mean directions. E is the expected early Eocene palaeofield. PEF is the present Earth field direction. Perimeter is the present horizontal.

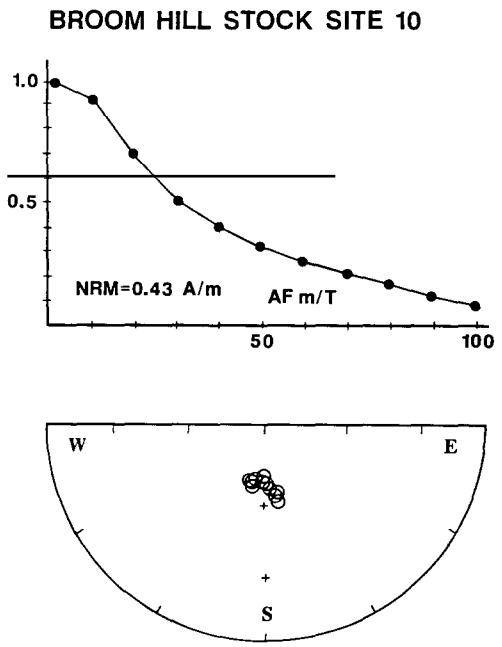


Fig. 11. Alternating field demagnetization of Broom Hill agamatite, site 10. See caption to Fig. 4.

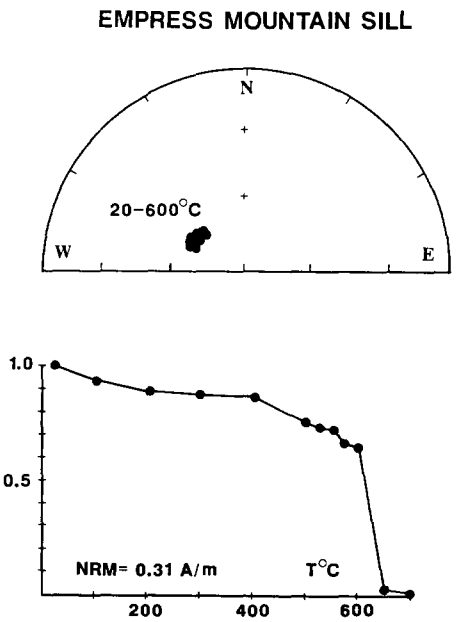


Fig. 13. Thermal demagnetization of Empress Mountain sill, site 7. See caption to Fig. 4.

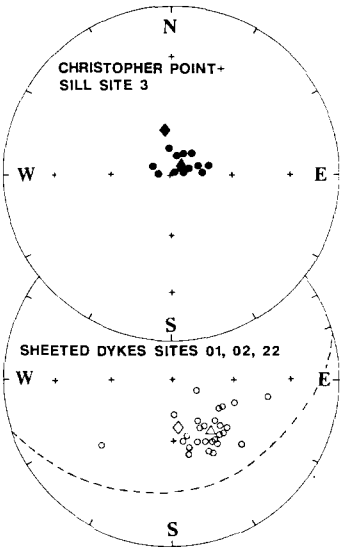


Fig. 14. Specimen directions in Christopher Point sill above and sheeted dykes sites 01, 02 and 22 below. Mean direction marked by triangles. E is the mean direction of the geomagnetic field (normal and reversed) in early Eocene expected in sampling area. Perimeter is the present horizontal. Dashed line in the lower diagram is plane of 'bedding'.

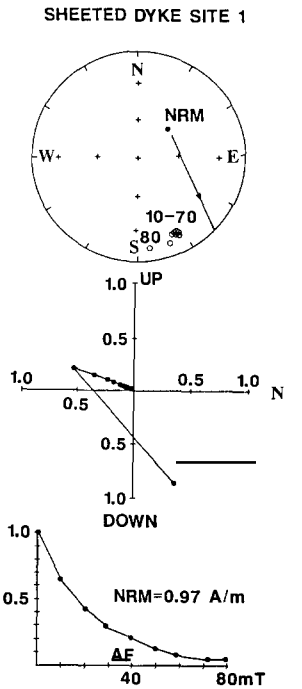


Fig. 15. Alternating field demagnetization of feldspar diabase from sheeted dykes, site 1. Orthogonal plot centre. See caption to Fig. 4. Directions relative to 'bedding'.

Empress Mountain sill, have very steep inclinations (Fig. 14).

4. Diabase

Diabase was sampled at Broom Hill and near Rocky Point, especially the latter. Alternating field and thermal demagnetization of Rocky Point diabase is shown in Figs. 15 and 16. Unlike the gabbros, viscous remanent magnetizations (VRM) acquired in the present Earth's field (PEF) are large. These are readily demagnetized, the directions becoming reversed at temperatures of 400 °C and a.f. of 10 mT. The inclinations are shallow relative to 'bedding' (Table 2).

The magnetizations of the Broom Hill diabase (site 22) are also dominated by soft present field VRMs, but small high coercive force magnetizations are present (Fig. 17). In situ the directions observed from sites 1, 2 and 22 (Fig. 3) are close

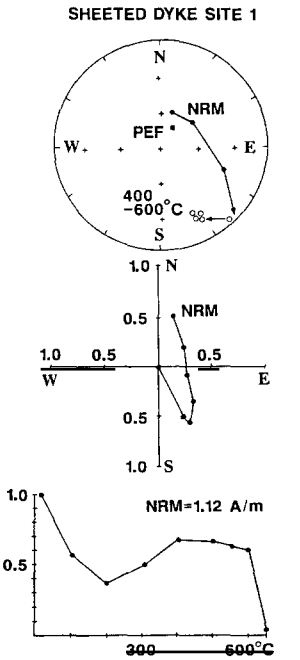


Fig. 16. Thermal demagnetization of feldspar porphyry from sheeted dykes at site 1. Directions relative to 'bedding'.

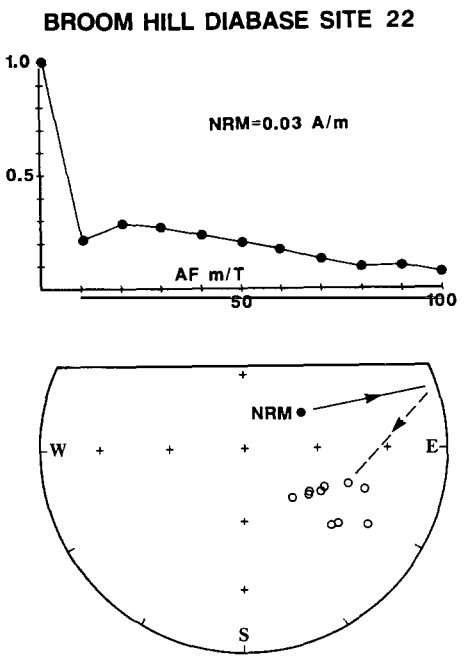


Fig. 17. Alternating field demagnetization of diabase from Broom Hill, site 22. See caption to Fig. 4. Perimeter is the present horizontal.

to the reversed magnetizations observed from the East Sooke gabbro (cf. Figs. 14 and 6).

5. Sooke Formation

Two units of the Sooke Formation were studied from the west coast of Becher Bay (Fig. 3). The sandstone is generally too friable to sample, but 16 cores were obtained from a harder ferruginous layer ~ 0.3 m thick. This layer immediately overlies a massive conglomerate containing boulders of up to 2 m in diameter. Sixteen boulders were sampled.

The remanent magnetization of the sandstone decays exponentially during a.f. demagnetization, remaining consistent in direction up to 100 mT when only a few per cent remains. The directions of Fig. 18 have been estimated by line-fitting (Kirschvink, 1980) in the range 10–70 mT. Thermal demagnetization was not possible, the specimens disintegrate. The mean direction in situ (D, I) 018°, 70° ($\alpha_{95} = 4°$) is statistically identical

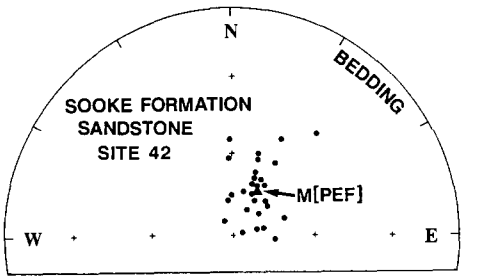


Fig. 18. Directions of magnetization in sandstone of the Sooke Formation, site 42. M is the mean direction and is identical to the present Earth's field (PEF).

with the present field (021°, 70°). The magnetization could, therefore, be contemporary and produced by seasonal redistribution of fine-grained haematite in these highly porous rocks. There is no reason to believe that it records the Oligocene field, although it could. The sandstones dip gently southeast (4° at 130°), and after correction, their mean magnetization direction is 028°, 71° (Table 1).

Boulders in the conglomerate are petrologically indistinguishable from the East Sooke stock they immediately overlie, and thermal and a.f. demagnetization data match those observed in situ. Within each boulder, directions agree very well (mean standard deviation of directions in speci-

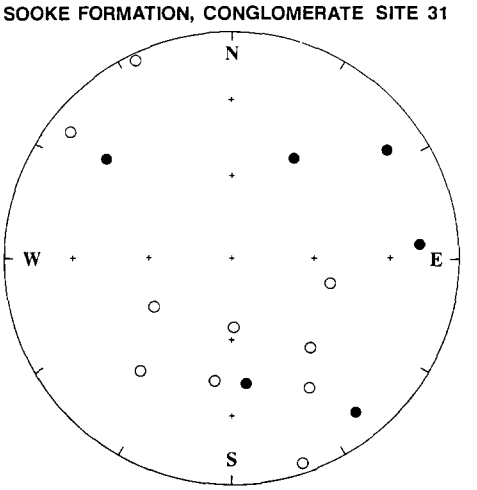


Fig. 19. Mean directions of magnetization in 16 boulders (two cores and four specimens per boulder) from base of Sooke Formation. Directions are random, $k = 1.2$. Perimeter is the present horizontal.

mens from the same boulder is 6°), but directions from boulder to boulder, are randomly distributed (Fig. 19). Hence the magnetization of the East Sooke gabbro pre-dates the deposition of the Sooke Formation, which is Oligocene in age.

6. Discussion

All rock-units have mean directions that differ from the expected Eocene field (Table 2, Fig. 20). Three have shallower inclinations and two (three if the gabbro sill of site 3 is included) have steeper inclinations. Clearly the departures cannot be attributed to any overall rotation or translation of the Metchosin Complex. Rather they indicate local tilting within the complex. The net tilts required are as follows: East Sooke stock 10° at 93°E (±5°); Broom Hill stock 5° at 28°E (±7°) not significant; Empress Mountain sill 11° at 13°E (±7°); and Rocky Point stock 31° at 214°E (±12°). There is no evidence of the systematic 60° clockwise rotations characteristic of Eocene rock-units in other parts of the Crescent Terrane. Since the Metchosin Complex has been underthrust beneath Wrangellia, it is, in a tectonic sense, an exceptional area of the Crescent Terrane. That its palaeomagnetic signature should be different, is not, therefore, surprising.

These tilts could have occurred as a result of extension tectonics at or near the ridge crest, or they could have been produced later as the Clallam syncline was formed and the Metchosin Complex thrust beneath Wrangellia (Fig. 3).

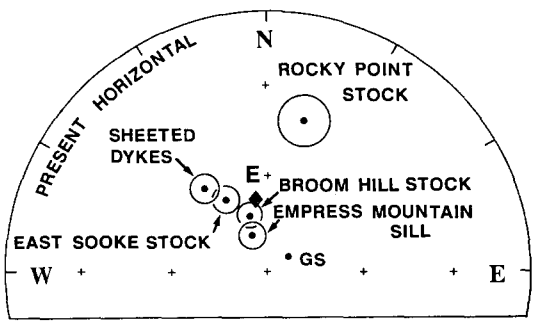


Fig. 20. Mean directions, irrespective of sign, in rock units of Metchosin Complex. Units represented by several sites have standard error circles ($P = 0.63$). GS is the gabbro sill at Christopher Point, site 3. All directions with respect to the present horizontal. E is the expected Eocene palaeofield.

The mean direction of the small stable magnetization of the sheeted dykes in situ is given in Fig. 20, and is not significantly different from that in the East Sooke stock in situ. Correction to bedding inferred from local mapping yields inclinations of 20–35° (Fig. 14 and Table 2). These correspond to a palaeolatitude of $< 20°$, far lower than can be contemplated for the Crescent Terrane during the Eocene (Beck, 1984). We infer, therefore, that the dykes were first tilted and afterwards acquired their present magnetization when the nearby East Sooke stock was intruded. The diabase was tilted and any primary magnetization they originally had was replaced by a secondary magnetization during intrusion of nearby gabbros. This magnetization (secondary in the diabase, primary in the gabbros), would be younger than the magnetization of the overlying upper crust.

The structure of the Metchosin Complex is not well-enough known to evaluate those palaeomagnetically observed tilts, but they are comparable in magnitude to dips observed in rare occurrences of bedded sequences. It is noteworthy that the Empress Mountain stock, closest to the Leach River fault has been tilted to the north roughly in the direction of the fault itself. Bodies further south have either negligible tilt (Broom Hill stock), have been tilted to the east-southeast (East Sooke stock and sheeted dykes), or have southerly tilts (Rocky Point stock) as the turn-over to the Clallam syncline is being approached (Fig. 3). Obviously, the picture is far from clear, but it is probable that these tilts were imposed during the emplacement of the Metchosin Complex in juxtaposition to Wrangellia. It ought to be possible to use the directions of magnetization in mid- and lower crustal rocks to further structural knowledge of such complex terranes.

At a time when virtually nothing was known about the magnetization of rocks of the oceanic crust, Vine and Matthews (1963) proposed that the source of marine magnetic anomalies was distributed throughout oceanic layers 2 and 3. The average magnetization required by their hypothesis was 2–5 $A\,m^{-1}$. Later, high magnetization (10 $A\,m^{-1}$) came to be observed from samples dredged from the young central parts of ocean ridges, and it became possible to explain magnetic anomalies

as originating in the upper few hundreds of metres, in the pillow basalts of layer 2a (Irving et al., 1970; Talwani et al., 1971). Later, as data has become available from DSDP drill core (e.g. Lowrie, 1977), and from ophiolitic sequences on land (e.g. Hall et al., 1987), it has become evident that other much deeper seated rocks within the oceanic crust are potential sources of marine magnetic anomalies. For example, diabasic gabbros of layer 2B (~ 3 A m^{-1}) and serpentinized peridotites of layer 4 of the upper mantle (~ 5 A m^{-1}) have strong stable magnetizations (Dunlop and Prevost, 1982). Somewhat surprisingly most studies of gabbros from layer 3 have yielded comparatively weak magnetizations (< 0.5 A m^{-1}, data summarized in table 7 of Dunlop and Prevost, 1982). Although this is true of many gabbro specimens from the Metchosin Complex, the Rocky Point stock for example, other bodies, notably the East Sooke stock and Empress Mountain sill, have high (2–3 A m^{-1}) and exceedingly stable magnetizations. The rocks are excellent recorders of the geomagnetic field, and they have retained this record faithfully during a not insignificant tectonic history. The susceptibilities of gabbros of the Metchosin Complex (Table 2) are such that Koenigsberger ratios (assuming a geomagnetic field of 0.05 mT) commonly exceed 5. Hence, in undisrupted oceanic crust such rocks would be expected to contribute substantially to marine magnetic anomalies.

In contrast, sheeted diabase of the Metchosin Complex, whose magnetization is predominantly a soft present field VRM, and the volcanics, which have scattered magnetizations of low intensity (0.1 A m^{-1}, Schwarz and Muller, 1982) could not have produced marine magnetic anomalies. Of the rocks of the Metchosin Complex, only gabbro stocks are likely to contribute. Our data are, therefore, consistent with the idea that, at least in some instances, marine magnetic anomalies have their origin in the lower part of the oceanic crust.

Acknowledgements

Laboratory measurements were made by Judith Baker. This work was carried out whilst N.W.D. Massey was a post-doctoral fellow at the Pacific Geoscience Centre.

Contribution of Geological Survey of Canada no. 48789. LITHOPROBE publication no. 52.

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